WORKED SOLUTION

STEP III 1987, Question 10

Pure • Bernoulli Polynomials

STEP III 1987, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s3-q10

The key idea

The defining rule fixes P_n only through its derivative, so the natural thing to induct on is the *derivative* of the difference $P_n(x+1) - P_n(x) - nx^{n-1}$: the inductive hypothesis kills it, leaving a constant, and the integral condition pins that constant down. After that, the sum is a telescoping sum, and for the cubes we only need P_4 up to its constant term, which cancels anyway.

Solution

The difference identity

For $n \geq 1$ let

$$Q_n(x) = P_n(x+1) - P_n(x) - nx^{n-1}.$$

We prove by induction that $Q_n(x) = 0$ for all x.

Base case $n = 1$. Since $P_1' = 1 \cdot P_0 = 1$, we have $P_1(x) = x + c$ for some constant c. Then $P_1(x+1) - P_1(x) = 1 = 1 \cdot x^0$, so $Q_1 \equiv 0$. (In fact $\int_0^1 (x+c)\,\mathrm{d}x = 0$ gives $c = -\frac{1}{2}$, but we don't need it.)

Inductive step. Suppose $n \geq 2$ and $Q_{n-1} \equiv 0$. Differentiating, and using $P_n' = nP_{n-1}$,

$$\begin{aligned} Q_n'(x) &= nP_{n-1}(x+1) - nP_{n-1}(x) - n(n-1)x^{n-2} \\ &= n[P_{n-1}(x+1) - P_{n-1}(x) - (n-1)x^{n-2}] \\ &= nQ_{n-1}(x) = 0. \end{aligned}$$

So Q_n is a constant, and it equals its value at $x = 0$. Because $n \geq 2$, the term nx^{n-1} vanishes at $x = 0$, so

$$Q_n(0) = P_n(1) - P_n(0).$$

Ask yourself How do we know $P_n(1) = P_n(0)$? The integral condition is about P_n, not its values.

Use the integral condition one level down. By the fundamental theorem of calculus,

$$P_n(1) - P_n(0) = \int_0^1 P_n'(x)\,\mathrm{d}x = n\int_0^1 P_{n-1}(x)\,\mathrm{d}x = 0,$$

since $n-1 \geq 1$, so the condition $\int_0^1 P_{n-1}\,\mathrm{d}x = 0$ applies. (This is exactly why the case $n = 1$ had to be done separately: $\int_0^1 P_0\,\mathrm{d}x = 1$.) Hence $Q_n \equiv 0$, which completes the induction:

$$P_n(x+1) - P_n(x) = nx^{n-1} \qquad \text{for all } n \geq 1.$$

The sum of powers

Put $x = m$ in the identity and add for $m = 0, 1, \ldots, k$. The right-hand side telescopes:

$$\begin{aligned} n\sum_{m=0}^{k} m^{n-1} &= \sum_{m=0}^{k} [P_n(m+1) - P_n(m)] \\ &= [P_n(1) - P_n(0)] + [P_n(2) - P_n(1)] + \cdots + [P_n(k+1) - P_n(k)] \\ &= P_n(k+1) - P_n(0), \end{aligned}$$

as required. (For $n = 1$ the $m = 0$ term is 0^0, which must be read as 1; both sides are then $k+1$.)

The sum of cubes

Take $n = 4$, so we need P_4. Working up from $P_0 = 1$, integrating each time and choosing the constant so that the integral over $[0,1]$ is zero:

$$\begin{aligned} P_1(x) &= x - \tfrac{1}{2}, \\ P_2(x) &= x^2 - x + \tfrac{1}{6} && \text{since } \textstyle\int_0^1 (x^2 - x)\,\mathrm{d}x = -\tfrac{1}{6}, \\ P_3(x) &= x^3 - \tfrac{3}{2}x^2 + \tfrac{1}{2}x && \text{since } \textstyle\int_0^1 (x^3 - \tfrac{3}{2}x^2 + \tfrac{1}{2}x)\,\mathrm{d}x = \tfrac{1}{4} - \tfrac{1}{2} + \tfrac{1}{4} = 0, \\ P_4(x) &= x^4 - 2x^3 + x^2 + c \end{aligned}$$

for some constant c. We don't need c: it cancels in $P_4(k+1) - P_4(0)$. Writing $y = k+1$,

$$P_4(y) - P_4(0) = y^4 - 2y^3 + y^2 = y^2(y-1)^2 = (k+1)^2k^2.$$

So the sum formula with $n = 4$ gives

$$\sum_{m=0}^{k} m^3 = \frac{k^2(k+1)^2}{4} = \left(\frac{k(k+1)}{2}\right)^2.$$

With $k = 1000$, $\frac{1}{2}k(k+1) = \frac{1}{2}\cdot 1000 \cdot 1001 = 500500$, so

$$\sum_{m=0}^{1000} m^3 = (500500)^2,$$

which is the required result.

Checks

- Small k: $k=2$ gives $0+1+8=9=\left(\frac{2\cdot3}{2}\right)^2$, and $k=3$ gives $36=6^2$.
- The same method with $n=2$ gives $2\sum m = P_2(k+1)-P_2(0)=(k+1)^2-(k+1)=k(k+1)$, the familiar triangular-number formula. Notice that the cube sum is the square of this, which is a nice thing to spot.
- By computer, building $P_1,\dots,P_{12}$ with exact fractions confirms the difference identity at random rational points and the sum formula for $k\le 39$. Summing the cubes directly gives $250\,500\,250\,000 = 500500^2$. (The constant we skipped turns out to be $c=-\frac{1}{30}$.)

Common slips

- Trying to fix the constant in Q_n by integrating Q_n over $[0,1]$. That brings in $\int_1^2 P_n$, which we know nothing about. Evaluating at $x=0$ and using the condition on P_{n-1} is much cleaner.
- Forgetting that the step $P_n(1)=P_n(0)$ needs $n\ge 2$, so the base case must be $n=1$ and checked by hand.
- Spending time finding the constant term of P_4, which cancels.

Beyond STEP

The Bernoulli polynomials have the generating function $\dfrac{te^{xt}}{e^t-1}=\sum_{n\ge0}P_n(x)\dfrac{t^n}{n!}$, and their values at 0 are the Bernoulli numbers $1,-\frac{1}{2},\frac{1}{6},0,-\frac{1}{30},\dots$. They are central to the Euler–Maclaurin formula, which compares a sum with an integral and its error terms, and they appear in Euler's evaluation of $\sum 1/n^{2k}$. For more on these ideas, try Graham, Knuth and Patashnik, *Concrete Mathematics*, which treats sums of powers and Bernoulli numbers in depth.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.